

A fuzz plugin and interactive resume. Each knob shapes the sound and opens a different part of the CV.

CV Fuzz

Checo Cadena
2026

Plugin

What it is

CV Fuzz is a fuzz plugin whose interface is Checo Cadena's resume. Each of its four knobs shapes the sound and opens one section of the CV.

The sound is a C++ model of a fuzz pedal Checo designed and built at NYU, with NYU grant funding. The signal goes through a bias stage that can be starved like a dying battery, up to +35.7 dB of drive, an asymmetric clipper that runs at 4 times the sample rate, a DC blocker, a tone control and an output level.

The interface is the CV. Turning a knob changes the sound and flips through one section: Fuzz opens Projects, Tone opens Experience, Level opens Education and Bias opens Recognition. The tick marks around each knob are detents, one per entry. The name is a pun: in modular synthesis, CV means control voltage, the signal that moves a parameter.

How it is made. CV Fuzz is built with JUCE 8 as an Audio Unit, a VST3 and a standalone app. The interface is a web page running inside the plugin window and bound to the real C++ parameters, so automation and saved sessions work as usual. The same page runs in a browser with a Web Audio approximation of the sound, which is the online demo.

Stage: version 1.0.1, released on GitHub on 28 September 2026 with installers for macOS, Windows and Linux. It is free and the source code is MIT licensed. The repository lists its own limits: the anti-aliasing filters are gentle, parameters are not smoothed yet, and the clipper is a tuned waveshaper, not a component-level model of the original pedal's circuit.

TO CONFIRM (Checo): the pedal. The repository says only that it was designed and built at NYU with NYU grant funding. The grant's name and year, the pedal's name, its circuit and controls, and a photo of it are not in the repository.

TO CONFIRM (Checo): press, users or performances with the plugin. None are in the repository, so none are claimed here.

Resume

Four knobs, four chapters

As the plugin shows it. Turning a knob steps through the entries of its section.

Fuzz	Tone	Level	Bias
Drive, 0 to +35.7 dB	500 Hz to 6 kHz	Output volume in dB	Starves the stage, 0 to 100%
Projects	Experience	Education	Recognition

Experience

Computer Control Engineer
Amorphic Robot Works, 2023 to now

Academic Technology Staff
Berklee NYC, 2026 to now

Project Manager
Aerial Entertainment, 2026 to now

Education

M.A., Creative Media and Technology
Berklee NYC, 2025 to 2026

Generative music agents
IRCAM, Paris, Feb 2025

B.S., Integrated Design and Media
NYU Tandon School of Engineering, 2020 to 2024

Recognition

Superstar EP
Checo Cadena, Jan 2025

Arts Incubator Fellow
NYU Production Lab, Oct 2024

Tidal Shift Award
Portland Museum of Art, Dec 2023

Projects

CV Fuzz
Plugin · 2026

ONE/EIGHT iPod Nano
Installation · 2026

Dual Pneuma
Robotic art · 2024

Liberation Through Amplification
Performance · 2024

Vida
Film score · 2024

Skills

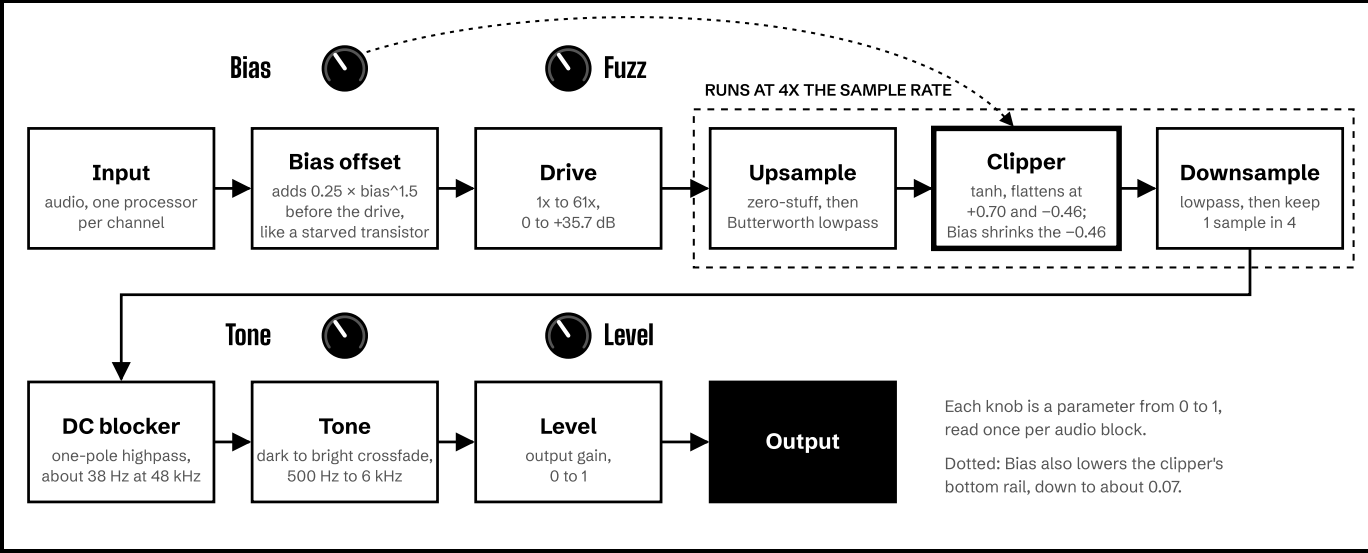
Code	C++, JUCE, Max/MSP, agentic coding
Audio	Ableton Live, Pro Tools, Logic Pro X
Sound	Spatial audio, synthesis, installations
Hardware	Pedal building, sensors, MIDI

Each entry opens a page with one plain sentence, a few labeled facts and links. The Resume PDF button opens a PDF of the resume bundled inside the plugin.

TO CONFIRM (Checo): the released build (v1.0.1) predates the last change on main, which rewrote the "My role" line of the Dual Pneuma entry. A new release is needed for the downloadable plugin to show the current text.

DSP

The signal path



Bias models a starved transistor. It adds an offset before the drive, so anything quieter than the offset sits on one side of the clipper and the DC blocker turns it into silence: tails choke off and loud hits sputter through. At full Bias the bottom rail also drops from 0.46 to about 0.07. On a synthesized drum loop at Fuzz 60%, 39% of the audible time chokes to near-silence at half Bias and 70% at full Bias.

The clipper uses tanh on each half of the wave, flattening at +0.70 on top and -0.46 below. A symmetric clipper makes only odd harmonics. The asymmetry adds even ones, which give fuzz its thicker sound. With a pure 220 Hz sine, the even harmonics sit about 35 dB below the fundamental.

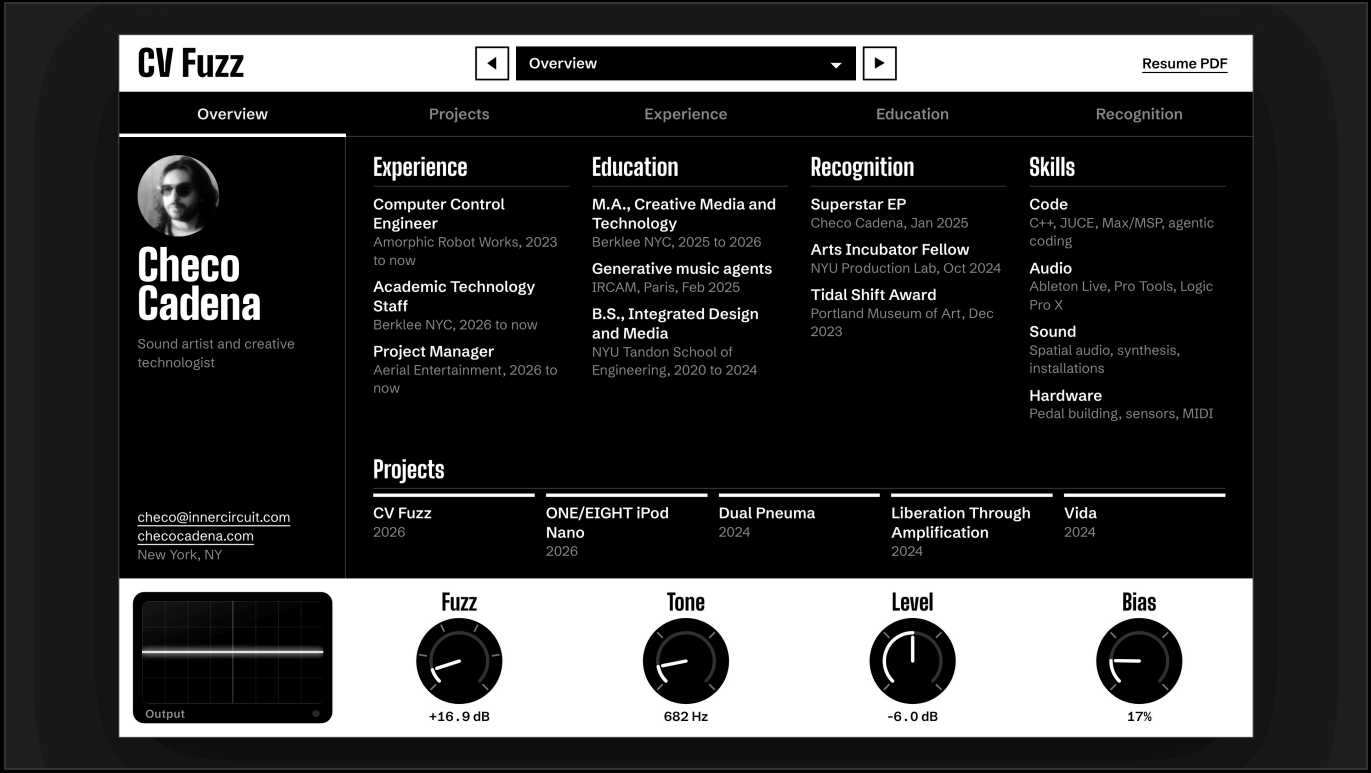
Oversampling. The clipper runs at 4 times the sample rate, between two second-order Butterworth lowpass filters at 0.45 times the original rate, to reduce aliasing.

Tone and Level. Tone crossfades between the signal and a lowpassed copy whose cutoff rises from 500 Hz to 6 kHz, a simple stand-in for a pedal's passive tone network. Level is the output gain.

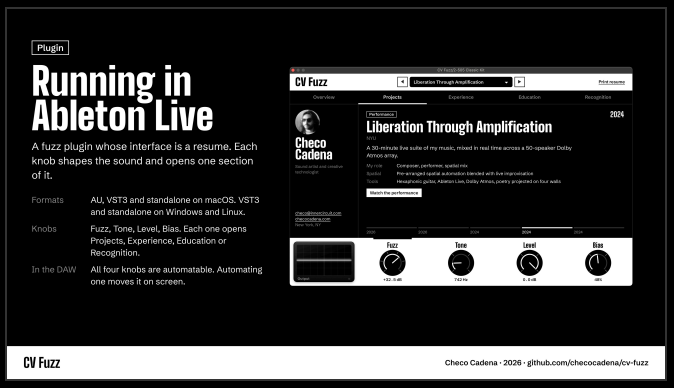
Architecture. The DSP is one C++ class with no framework dependency. The interface is one HTML file compiled into the plugin, so it never needs network access. A lock-free ring buffer of 4,096 samples feeds the oscilloscope on every display refresh.

Tested with a standalone C++ harness that renders a pure sine, a synthesized plucked-string riff, a sine sweep from 80 Hz to 2 kHz and a synthesized drum loop, analyzed in Python. **Next steps listed in the repository:** steeper anti-aliasing, parameter smoothing, a DC blocker that follows the sample rate, a circuit-level model of the original pedal, and an RNBO version.

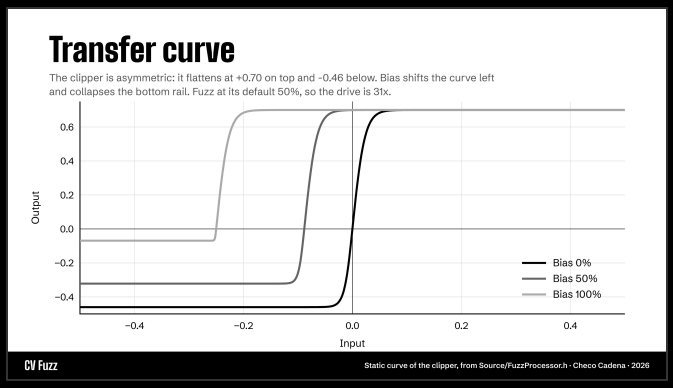
Images



1 · The interface. The plugin's own interface file (Source/ui/index.html), rendered at high resolution on its Overview page.



2 · In Ableton Live. The repository's screenshot (docs/media), shown at its native resolution. It was taken before the "Print resume" button was renamed "Resume PDF".



3 · Transfer curve. Redrawn from the formulas in Source/FuzzProcessor.h, after the repository's plot.

Supplied as 3840 × 2160 JPEGs with this kit. Images 1 and 2 are real captures of the plugin; image 3 is a chart computed from its code.

Setup and links

Version	1.0.1, released 28 September 2026. Download from `github.com/checocadena/cv-fuzz/releases/latest`
macOS	macOS 11 or later, Apple Silicon and Intel. `CV-Fuzz-macOS.pkg`, signed and notarized. Audio Unit, VST3 and standalone app; the installer lets you pick which.
Windows	Windows 10 or 11, 64-bit. `CV-Fuzz-Windows-Setup.exe`: VST3 and standalone app. The installer is not code-signed, so SmartScreen may warn: choose More info, then Run anyway.
Linux	x86-64. `CV-Fuzz-Linux-x64.tar.gz`, then run `install.sh`: VST3 in `~/.vst3`, standalone as `cv-fuzz`. Needs WebKitGTK 4.1.
In a DAW	In Ableton Live: Settings > Plug-Ins, turn on Audio Units (Mac) and VST3, then Rescan. CV Fuzz is under Checo Cadena.
From source	`./build-mac.sh` on a Mac with Xcode Command Line Tools and CMake. JUCE 8.0.4 is downloaded automatically. Windows and Linux builds run on GitHub Actions.
License	The source code is under the MIT License, copyright 2026 Sergio Cadena Gonzalez (Checo Cadena). It does not cover the resume content, photographs or the resume PDF, which remain the property of Checo Cadena. JUCE is licensed separately under its own terms.

TO CONFIRM (Checo): the Windows and Linux builds come from GitHub Actions. The repository does not say whether they were tried on a Windows or Linux machine.

Website: `checocadena.github.io/cv-fuzz` · **Try it in a browser:** `checocadena.github.io/cv-fuzz/demo`
Code: `github.com/checocadena/cv-fuzz` · **Demo video:** `docs/media/cv-fuzz-demo.mp4` in the repository (35 seconds)

About the artist

Checo Cadena is a New York-based singer-songwriter, sound artist and creative technologist from Monterrey, Mexico, who writes and produces records and builds the machines around them: plugins, robots, installations and software. Since 2023 Cadena has been Computer Control Engineer at Amorphic Robot Works, Chico MacMurtrie's robotic sculpture studio. Cadena holds a B.S. in Integrated Design and Media from NYU Tandon and an M.A. in Creative Media and Technology from Berklee NYC.

Contact: Checo Cadena · `github.com/checocadena` · `checocadena.com`